

# Course Introduction

CSCI 1101

2024-01-23

## About the Course

# Course Objectives

- Learn to write, read, and reason about computer programs.
- Use programming to solve interesting problems:
  - Decompose complex problems into simpler subproblems,
  - solve each subproblem,
  - combine the solutions to solve the original problem.

# Course Components

## Lectures:

introduce new concepts together with examples

## Activities:

in-class collaborative hands-on programming practice

## Homework Assignments:

targeted programming problems with automated feedback from CodeRunner

## Projects:

larger programming problems requiring creativity and problem-solving

## Exams:

assess mastery of the course material

# Evaluation

Course grades will be computed as follows:

|                      |     |
|----------------------|-----|
| homework assignments | 20% |
| projects             | 20% |
| midterm exam 1       | 15% |
| midterm exam 2       | 20% |
| final exam           | 25% |

You get 5 free *late days* to use as you like on homeworks and projects.

In order to use a late day on an assignment you must notify the instructor at least six hours before the deadline to ensure time to enter the extension in CodeRunner.

# About Programming

# Discussion

What do computers do?

# Discussion

What is an algorithm?



# Discussion

How do we communicate an algorithm to a computer?

# Programming Languages

A **programming language** is a formal language for for representing and manipulating information, and performing tasks.

A **imperative** programming language does this using sequences of instructions (do this, then do that).

In this course we will use an imperative programming language called **Python**.

# Why Learn to Program?

- Interact with computers on your own terms.
- Automate laborious tasks.
- Bring your ideas to life.
- Gain a mental force multiplier.
- It's fun!

What's your reason?

# How to Learn to Program

learn *what*:

concepts (lecture, reading)

learn *how*:

skills (activities, homework, projects)

Learning to program is a process,  
to be successful, you need to invest time and attention.

# Programming is like...

## a natural language:

It has a *vocabulary* and rules of *grammar* that we need to learn and follow.

## a natural science:

To understand how a piece of software works we can perform *observations* and *experiments*.

## a mathematical theory:

Computers operate by following precise *rules*.

By learning these rules we can predict and direct the computer's *behavior*.

# Why Might Programming Seem Hard?

- Unfamiliar syntax
- High level of precision and detail required

# Language and Syntax

**Syntax** is the rules for expressing an idea in a particular language.

Natural languages tend to be complex and flexible.

*To die, to sleep; to sleep, perchance to dream; aye, there's the rub,  
for in that sleep of death, what dreams may come, when we have  
shuffled off this mortal coil, must give us pause.*

Formal languages tend to be simple, but rigid.

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

# Algorithms

Named for Al-Khwārizmī, a 9th c. Persian mathematician and astronomer.

An **algorithm** is a complete description of a procedure for accomplishing a task.

- must be clear and unambiguous
- no ingenuity or creativity required
- must handle every possible case
- must eventually produce a result

Programming languages let us communicate algorithms to computers.

Then the computers can do the work of carrying them out!



# Computers

These days our computers are electronic devices,  
But originally, a “computer” was a person.



# About Python

# Python

**Python** is:

- a popular high-level imperative programming language,
- a collection of software tools to write and run programs in that language.

In this course we will use the Python language version 3.

# Python, the Language

## popular

Lots of users, platforms, libraries and tools

## high-level

Abstracts away details of the underlying hardware, operating system, etc.

# Python, the Platform

- An open-source project (`http://python.org`)
- A program interpreter (`python`, `python3`)
- A program editor (`idle`, `idle3`)
- Documentation (books, videos, reference manuals, etc.)
- Community (mailing lists, blogs, conferences, etc.)

# Installing Python

To install Python on your computer:

- browse to [python.org](https://python.org),
- choose “Downloads”,
- download the default version for your system,
- launch the installer and follow the steps.

Python comes with a built-in program editor called **IDLE**.

Optionally, you may also install another program editor called **Wing 101**.

You can find this at: [wingware.com/downloads/wingide-101](https://wingware.com/downloads/wingide-101).

# Assignments

## To Do:

Before our next meeting, try to do the following:

- Install Python (and optionally Wing 101) on your computer
- Join the Slack workspace
- Sign into CodeRunner (if it asks for a code, use “2024SP1101”)

If you encounter any problems, ask for help.

Before next Monday complete the following tasks on CodeRunner:

- *Homework on Collaboration Policy*
- *Homework 0: Intro to CodeRunner*